Evaluating the Efficiency of Ion vs. Chemical Propulsion in Deep Space Trajectories

A Comparative Data Analysis Using NASA Mission Data and Orbital Mechanics

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Summary

Choosing the right propulsion system is one of the biggest decisions in planning a space mission. The right propulsion system decides how much of a spacecraft's weight can go into instruments and cargo instead of fuel, and how long the trip will take. This paper compares two propulsion types, ion propulsion and chemical propulsion, using NASA engine data and standard orbital mechanics equations. The comparison is applied to two example missions: a trip to Mars and a trip to Jupiter. By calculating fuel needed, thrust-to-weight ratio, and travel time for each propulsion type, this paper shows when each type works the best.

Introduction

As space missions travel farther and face newer difficulties, choosing between chemical and ion propulsion has become one of the most important engineering decisions in mission planning. Chemical propulsions is the technology most commonly used in almost every rocket launch and many orbit-insertion burns. It creates a lot of thrust for a short time, but it's not very fuel-efficient. Ion propulsion on the contrary is far more fuel-efficient; first tested on NASA’s Deep Space 1 mission and later used on the Dawn mission. Although it is more fuel-efficient, it only produces a tiny amount of thrust. Because of this, it has to run continuously for months or years to build up speed.

The point of this paper is to find out when ion propulsion works better than chemical propulsion for a trip through space, and what exactly do we gain or lose by choosing one over the other. Mars and Jupiter were picked as the two example missions because they require very different amounts of speed change, and because NASA has flown real

missions to both using each propulsion type. This makes it possible to check the numbers in this paper against real mission results.

Background

Chemical Propulsion

Chemical Propulsion works by burning fuel and an oxidizer together and pushing the hot gas out through a nozzle. It is still the standard choice for launching rockets and for orbit-insertion burns, which need a lot of thrust delivered quickly. Its main weakness however, is a low specific impulse (Isp), which is the standard way of measuring how efficiently an engine turns fuel into thrust. Engines that burn fuel for orbit-insertion burns usually have an Isp of around 317 seconds [2].

NASA's Juno spacecraft is a good example. Its main engine, a 645-newton Leros-1B engine burning hydrazine and nitrogen tetroxide, fired for just 35 minutes to slow the spacecraft down by 542 m/s and let it get captured into orbit around Jupiter [5][6]. Juno did not fly straight to Jupiter using only its engine; however, it used a flyby of Earth in October 2013 to gain extra speed for free, since no rocket available at the time was powerful enough to send it there directly [9]. Section 4 shows with real numbers why a direct flight without that flyby would not have worked.

Ion (Electric) Propulsion

Ion propulsion works very differently. It turns a propellant into charged particles (ions) and pushes them out using an electric field. This gives it a very high specific impulse. For example NASA's NEXT engine reaches up to 4,190 seconds at full power, about thirteen times higher than a typical chemical engine [2]. Although NEXT produces a very high specific impulse, it only produces 236 millinewtons of thrust, which is a thousand times weaker than a chemical engine [2]. This means an ion-powered spacecraft speeds up very slowly, and has to keep its engine running for months or years to get anywhere.

NASA's Dawn mission shows this trade-off. Dawn used an older, even weaker ion engine called NSTAR (about 92 millinewtons of thrust). It only needed 425kg of xenon fuel, but it had to fire its engines for about 2,000 days total to build up roughly 11 km/s of speed change [3][4].

The Core Trade-off

Chemical and ion engines are almost opposites; chemical engines are fast but burn through a lot of fuel, whereas ion engines barely use any fuel but take a long time. Mission planners have to decide whether speed or fueling savings matter more for their mission. Section 4 puts real numbers to this trade-off.

Methodology

This paper uses two equations from orbital mechanics, along with NASA engine specifications and mission data, to compare the two propulsion types.

The Hohmann transfer equations calculate the speed change (delta-v) and travel time needed for the most fuel-efficient path between two circular orbits; one engine burn to leave the starting orbit, a coast along a curved path, and one more burn to settle into the new orbit. These are equations taught in orbital mechanics [7]. They are used here to find the delta-v and travel time for a chemical-propulsion trip from Earth's orbit to Mars's orbit and to Jupiter's orbit.

The rocket equation — propellant mass = dry mass x (e^(delta-v / Isp x g0)) - 1) — calculates how much propellant is needed to reach a certain delta-v, based on an engine's Isp [8]. This equation is applied to the delta-v numbers from the Hohmann transfer calculation, once using a chemical engine's Isp and once using an ion engine's Isp, so the fuel needed by each type can be compared directly from the same trip.

All engine numbers used (Leros-1b and NEXT) come from flown NASA hardware [2][6]. All orbital distances used (Earth: 1.000 AU, Mars: 1.524 AU, Jupiter: 5.204 AU) are each planet's average distance from the sun. The results are checked against mission data from Dawn and Juno and other published Hohmann transfer numbers to make sure they are accurate.

Results

Delta-v and Transfer Time (Hohmann Transfer)

Transfer	Total Delta-v required	Transfer time
Earth → Mars	5,596 m/s	259 days
Earth → Jupiter	14,437 m/s	998 days

The Mars number closely matches a number from an outside source; a University of Wisconsin orbital mechanics reference reports a minimum Earth → Mars transfer delta-v of 5.6 km/s, which is close to the number calculated in the chart [7]. Published data for Earth → Jupiter delta-v numbers vary more from around 11 to over 19 km/s. This is because the numbers depend on whether you start counting from Earth's orbit around the Sun or from low-Earth orbit, and whether the planets' orbits are treated as perfectly circular. The 14,437 m/s number used in this paper assumes departure from Earth's orbit around the Sun with both orbits treated as circles.

Propellant Mass (Rocket Equation)

Using the rocket equation with the delta-v numbers above, for a spacecraft with 1,000 kg of dry mass not counting fuel:

Transfer	Engine	Isp	Propellant Mass	Total mass with fuel	Percent of mass that is fuel
Earth → Mars	Chemical (Leros-1b class)	317s	5,050 kg	6,050 kg	83.5%
Earth → Mars	Ion (NEXT class)	4,190s	146 kg	1,146 kg	12.7%
Earth → Jupiter	Chemical (Leros-1b class)	317s	102,955 kg	103,995 kg	99.0%

Earth → Jupiter	Ion (NEXT class)	4190s	421 kg	1,421 kg	29.6%

The Jupiter chemical result is the most important number in this paper; flying straight to Jupiter with only a chemical engine would need over 100 times the spacecraft's own weight in fuel. This is not realistic to launch. This shows why real chemical-propulsion missions to Jupiter, like Juno, don't fly there directly but rather they use a gravity-assist flyby to gain speed for free instead of burning that much fuel.

Thrust-to-Weight Ration

Case	Thrust-to-weight ratio
Chemical, Mars mission	0.0184
Chemical, Jupiter mission	0.0097
Ion, Mars mission	0.0000201
Ion, Jupiter mission	0.0000185

The chemical engine’s thrust-to-weight ratio is roughly 900 to 1,000 times higher than the ion engine’s in both cases. This is the direct reason a chemical burn is over in minutes, while an ion transfer takes almost a year or longer.

Discussion

These results confirm and put real numbers behind the trade-off described in the background. For a Mars trip, chemical propulsion uses more fuel than ion propulsion (83.5% vs. 12.7% of total mass) but arrives faster in this case. That makes chemical propulsion a reasonable choice for a Mars mission where speed or a tight schedule matters, such as a crewed mission or a time-sensitive robotic one.

For Jupiter, the difference is much bigger: chemical propulsion needs a nearly impossible 99% of the spacecraft's mass in fuel, while ion propulsion only needs about 30%. This makes a strong case for either using ion propulsion or using chemical propulsion.

Overall, these results show that the fuel-savings advantage of ion propulsion over chemical propulsion grows quickly as a mission requires more delta-v. This makes ion propulsion more and more worth it the farther and more demanding the destination, up to the point where chemical propulsion alone stops being realistic without a gravity assist.

Limitations

- The Hohmann transfer calculations treat Earth, Mars, and Jupiter as perfect circles moving in the same plane, and do not account for the planets' real positions on a specific launch date.
- The rocket equation calculations assume one instant burn. In reality, an ion engine has to burn continuously for a long time, which adds some extra losses that a single quick burn does not have. This means the real ion fuel numbers could be a bit higher than shown here.
- This paper does not include the weight of the equipment an ion engine needs to work, like solar panels and power systems. Adding that weight would reduce ion propulsion's fuel advantage slightly.
- Gravity-assist trajectories, which real chemical-propulsion missions use to reach Jupiter, are not calculated here. The Jupiter chemical numbers in this paper are for a direct flight only, included on purpose to show why missions do not fly there directly.
- Published Earth-Jupiter delta-v numbers differ depending on starting assumptions (see results). The number used here follows one clearly stated, standard method, not a single number that everyone agrees on.

Conclusions

This paper compared chemical and ion propulsion for trips to Mars and Jupiter using standard orbital mechanics equations and NASA engine data (Leros-1b and NEXT), with the Mars delta-v number backed up by an outside source. The results show that chemical propulsion is still a workable, somewhat faster choice for Mars, while a direct chemical flight to Jupiter uses too much fuel to be realistic; which explains, with actual numbers, why real Jupiter missions use gravity assists. Ion propulsion offers a fuel-saving

advantage that grows bigger as a mission needs more delta-v, but it comes at the cost of a trip which takes years rather than months. In the future, this research can be improved by including gravity-assist paths for the chemical case, adding the weight of the ion engine's power system, and accounting for how an ion engine's thrust changes as it gets farther from the Sun.

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